P1:

i) $M = \min\{Y, X_1, \dots, X_n\}$, $M \sim Exp(n\theta + \mu)$

$$E(M^{-\frac{1}{2}}) = \int m^{-\frac{1}{2}} (n\theta+\mu) e^{-(n\theta+\mu)m} dm$$

for $X \sim Exp(\lambda)$, $E[X^n] = \int x^n \lambda e^{-\lambda x} dx$

let $u = \lambda x$ $du = \lambda dx$

$$E[X^n] = \int \left(\frac{y}{\lambda}\right)^n e^{-u} du = \frac{1}{\lambda^n} \int u^n e^{-u} du$$

$$= \lambda^{-n} \Gamma(1+n)$$

when $\lambda = n\theta+\mu$, $n = -\frac{1}{2}$

$\Gamma(1-\frac{1}{2}) = \Gamma(\frac{1}{2}) = \sqrt{\pi}$

Thus, $E(M^{-\frac{1}{2}}) = (n\theta+\mu)^{1/2}\sqrt{\pi}$

ii) $M = \max Y, X_1, \dots X_n$.

$$F(m) = P(M \le m) = \prod_{i=1}^{n} P(X_i < m) \cdot P(Y < m) = \prod_{i=1}^{n} (1-e^{-\theta m}) \cdot (1-e^{-\mu m})$$

$$= (1-e^{-\theta m})^n (1-e^{-\mu m})$$

$P(M > m) = 1 - P(M \le m)$ $u = t$ $v = f(t)$ $u' = 1$ $v = f(t)$

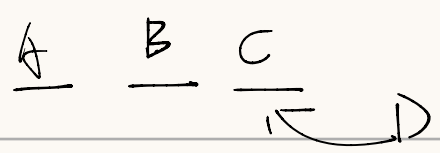
$$E(M) = \int m P(M=m) dm = m F(m) \Big|_0^\infty - \int F(m) dm$$

$$= -\int F(m) dm = \int P(M > m) dm$$

$$= \int 1 - (1-e^{-\theta m})^n (1-e^{-\mu m}) dm$$

P2.

$$P = \frac{n\theta}{n\theta+\mu} \times \frac{(n-1)\theta}{(n-1)\theta+\mu} \times \dots \times \frac{\theta}{\theta+\mu}$$



P3. a) If service time is exactly ten minutes and A, B, C started simultaneously, A, B must left when D start service. $P = 0$

b) $P(T_C < \min(T_A, T_B)) = \frac{\mu}{3\mu} = \frac{1}{3}$

After D started service, $T_D, T_A', T_B' \sim Exp(\mu)$

$P(T_D < \min(T_A', T_B')) = \frac{1}{3}$

Total $P = (\frac{1}{3})^2$

$Exp(7)$

P4.

a). T: time of next train arrive.

$X \sim$ Poisson $(.7T)$ $\quad T \sim$ Uniform $(0,1)$

$E(X) = E(E(X|T)) = E(7T) = 7E(T) = \frac{7}{2}$

b) $Var(X) = Var(E[X|T]) + E[Var(X|T)]$

$= Var(7T) + E(7T)$

$= \frac{91}{12}$

P5.

a) $N_i(t)$: # of type i car have passed by time t

$N_i(t) \sim$ Poisson $(\lambda t \cdot p_i)$

$P(N_1, N_2, N_3 \geq 1, N_4 = 0) = e^{-\lambda t p_4}(1 - e^{-\lambda t p_3})(1 - e^{-\lambda t p_2})(1 - e^{-\lambda t p_1})$

b) $P(N_1 + N_2 = 6) = \sum_{i=0}^{6} P(N_1 = i) P(N_2 = 6 - i)$

$P = \frac{P(N_1 = 4, N_2 = 2)}{P(N_1 + N_2 = 6)} = \frac{e^{-\lambda t p_1} \frac{(\lambda t p_1)^4}{4!} e^{-\lambda t p_2} \frac{(\lambda t p_2)^2}{2!}}{\sum_{i=0}^{6} e^{-\lambda t p_1} \frac{(\lambda t p_1)^i}{i!} e^{-\lambda t p_2} \frac{(\lambda t p_2)^{6-i}}{(6-i)!}}$